Examiner only

3. (a) **A**, **B**, **C** and **D** are equations for reactions used to make salts.

(i) Complete the equation for each reaction. [4]

A $Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + $ .... (g)

B $AgNO_3(aq) + NaCl(aq) \longrightarrow AgCl(s) + $ .... (aq)

C $ZnO(s) + 2HNO_3(aq) \longrightarrow$ .... $(aq) + H_2O(l)$

D $CuCO_3(s) + $ .... $(aq) \longrightarrow CuSO_4(aq) + H_2O(l) + CO_2(g)$

(ii) Outline the **three** steps you would carry out to get a **pure** sample of silver chloride, AgCl, from the reaction mixture in equation **B**. [2]

....

....

....

....

3410UD01 11

Examiner only

(b) Sodium chloride is made when sodium hydroxide reacts with hydrochloric acid.

sodium hydroxide + hydrochloric acid ⟶ sodium chloride + water

(i) Write a balanced equation for the reaction. [2]

....

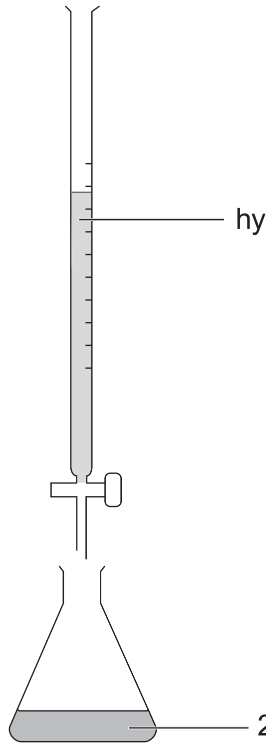
(ii) A student was asked to prepare pure sodium chloride crystals using $25.0\,cm^3$ of sodium hydroxide solution.

The flow diagram shows the student's plan.

Stage **1**		Stage **2**		Stage **3**
Carry out a titration to find the exact volume of acid needed to neutralise all of the alkali	→	Make a pure sodium chloride solution	→	Obtain sodium chloride crystals

Examiner only

I. The student sets up the apparatus below to carry out stage **1**.



Describe how the student will find the exact volume of acid needed to neutralise all of the alkali. [2]

....

....

II. The neutral solution obtained in stage **1** is not pure.

Describe what the student must do in stage **2** to make a pure sodium chloride solution. [2]

....

....

....

III. State what the student must do in stage **3** to obtain sodium chloride crystals. [1]

....

....

3410UD01 13

13